

Computer Forensics



Unit – 3

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Unit – 3

Windows Forensics



New Technology File System



NTFS



- Introduction to NTFS
- What is MFT?
- Structure of MFT
- Benefits of MFT
- Drawbacks of MFT
- MFT Records
- NTFS Forensics



NTFS: Introduction



- NTFS (New Technology File System) is a file system used by the Windows operating system to store and manage files on a storage device.
- It was introduced with Windows NT 3.1 in 1993 and has since become the default file system for Windows NT-based operating systems, including Windows 2000, XP, Vista, 7, 8, and 10.



NTFS: Introduction



- NTFS is designed to provide improved performance, reliability, and security compared to the older FAT (File Allocation Table) file system used by Windows 9x and earlier versions of Windows NT.
- NTFS supports features such as file and folder permissions, encryption, compression, and disk quotas, which are not available in FAT.



NTFS: Introduction



- NTFS also has a more advanced structure for organizing files and directories, including the Master File Table (MFT), which stores information about each file and directory on the volume.
- NTFS has a maximum file size of 16 exabytes (EB) and a maximum volume size of 256 terabytes (TB), although the practical limits may be lower depending on the operating system and hardware.



NTFS: Introduction



- NTFS is commonly used on hard disk drives (HDDs), solid-state drives (SSDs), and other storage devices, both internal and external.
- NTFS has several features that make it a popular choice for Windows users. Some of these features include:
 - Support for larger file sizes and volumes compared to FAT
 - Improved security through file and folder permissions, encryption, and access control



NTFS: Introduction



- NTFS has several features that make it a popular choice for Windows users. Some of these features include:
 - Support for journaling, which helps prevent data loss in case of power failures or system crashes
 - Compression and encryption capabilities
 - Support for symbolic links and hard links
 - Built-in support for file compression and decompression



NTFS: Introduction to MFT



- MFT is a database that stores information about every file and directory on an NTFS volume.
- It is located at a fixed location on the disk and is allocated a fixed amount of space.
- MFT is created when the volume is formatted with NTFS and grows as more files and directories are added to the volume.



NTFS: MFT Structure



- MFT is divided into fixed-size records, each of which represents a file or directory.
- The first record is called the Master File Record (MFR) and contains information about the MFT itself.
- Each record contains attributes that describe the file or directory, such as its name, size, permissions, and timestamps.



NTFS: MFT Records



- Attributes can be resident (stored entirely within the MFT record) or non-resident (stored outside the MFT record).
- Attributes can also be standard (defined by NTFS) or non-standard (defined by applications or third-party software).
- The attributes within an MFT record are structured in a header and a body.
- The header contains metadata about the attribute, such as its type, length, and flags. The body contains the actual data of the attribute.



NTFS: MFT Structure



- This header contains information about the MFT entry, such as its attributes and the location of the next entry.
- The actual size of the MFT entry, which includes the header and its attributes, can vary depending on the file's characteristics and the size of its attributes.



NTFS: MFT Records



- MFT records are the building blocks of the Master File Table (MFT) in NTFS file system.
- Each MFT record represents a file or directory on the volume.
- The first MFT record is the Master File Record (MFR), which contains information about the MFT itself.



NTFS: MFT Records



- Each MFT record contains a unique identifier called the File Reference Number (FRN), which is used to reference the record from other MFT records.
- MFT records contain attributes that describe the file or directory, such as its name, size, permissions, and timestamps.



NTFS: MFT Records



- Microsoft reserves the first 16 MFT entries for file system metadata files.
- Every file system metadata file is listed in the root directory, although they are typically hidden from most users.
- The name of each file system metadata file begins with a “\$,” and the first letter is capitalized.



NTFS: MFT Records



- MFT records can contain references to other MFT records. For example, a directory MFT record contains references to the MFT records of the files and subdirectories within that directory.
- MFT records can also contain unused space, which can be used to store additional attributes or data in the future.



NTFS: MFT Records



- The size of an MFT record is determined by the cluster size of the volume.
- The default cluster size is 4KB, so the size of an MFT record is usually a multiple of 4KB.
- However, the size can be increased or decreased depending on the needs of the system.



NTFS: MFT Records



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NTFS: MFT Entry Records



- The default MFT entry records are the attributes that are created by default for every file or directory in the NTFS file system.
- These attributes provide essential information about the file or directory and help to organize and manage the data on the volume.



NTFS: MFT Entry Records



- Every MFT entry also has a 16-bit sequence number that is incremented when the entry is allocated.
- The MFT entry and sequence number are combined, with the sequence number in the upper 16-bits, to form a 64-bit file reference address.
- NTFS uses the file reference address to refer to MFT entries because the sequence number makes it easier to determine when the file system is in a corrupt state.



NTFS: MFT Entry Records



- The default MFT entry records include the following attributes:
 - \$STANDARD_INFORMATION,
 - \$FILE_NAME,
 - \$OBJECT_ID,
 - \$SECURITY_DESCRIPTOR,
 - \$VOLUME_NAME,
 - \$VOLUME_INFORMATION, and
 - \$DATA.



NTFS: MFT Entry Records



- The `$STANDARD_INFORMATION` attribute contains information about the file or directory, such as its timestamps (creation, modification, and access) and permissions (read, write, execute).
- The `$FILE_NAME` attribute contains the name of the file or directory, as well as information about its parent directory, creation time, and other attributes.



NTFS: MFT Entry Records



- The \$OBJECT_ID attribute contains a unique identifier for the file or directory, which can be used to track it across different volumes.
- The \$SECURITY_DESCRIPTOR attribute contains information about the security permissions for the file or directory, such as the owner, group, and access control lists.



NTFS: MFT Entry Records



- The \$VOLUME_NAME attribute contains the name of the volume, which is used to identify the volume and differentiate it from other volumes.
- The \$VOLUME_INFORMATION attribute contains information about the volume, such as its creation time, serial number, and number of sectors.



NTFS: MFT Entry Records



- The \$DATA attribute contains the actual data of the file or directory, such as the contents of a text file or the binary code of an executable file.
- The \$DATA attribute can be resident (stored entirely within the MFT record) or non-resident (stored outside the MFT record), depending on the size of the file or directory.



NTFS: MFT Entry Records



- In addition to these default MFT entry records, there are many other attributes that can be added to an MFT record, depending on the needs of the system and the applications that use it.
- These non-standard attributes are defined by applications or third-party software and are not part of the NTFS specification.



NTFS: Additional MFT Entry Records



- \$MFT (File Record 0): Contains information about the MFT itself and serves as the root of the file system.
- \$MFTMirr (File Record 1): A backup copy of the MFT in case the original becomes corrupted.
- \$LogFile (File Record 2): Contains the NTFS transaction log, which records changes to the file system.



NTFS: Additional MFT Entry Records



- \$Volume (File Record 3): Contains information about the volume, such as the volume label and serial number.
- \$AttrDef (File Record 4): Contains definitions for all possible attributes that can be associated with files or directories.
- \$Bitmap (File Record 6): Contains a bitmap of used and unused clusters on the volume.



NTFS: Additional MFT Entry Records



- \$Boot (File Record 7): Contains the boot sector of the volume.
- \$BadClus (File Record 8): Contains a list of bad clusters on the volume.
- \$Secure (File Record 9): Contains information about user accounts and security policies.



NTFS: Additional MFT Entry Records



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- \$BadClus (File Record 8): Contains a list of bad clusters on the volume.
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NTFS: Benefits of MFT



- MFT provides fast access to file and directory information because it is located in a fixed location on the disk.
- It reduces fragmentation because file and directory information is stored in a contiguous block of space.
- MFT allows for advanced features such as file compression, encryption, and security descriptors.



NTFS: Drawbacks of MFT



- MFT can become fragmented, which can slow down access to file and directory information.
- MFT can become corrupted, which can result in data loss or system instability.
- MFT is not compatible with other operating systems, which can cause problems when sharing data between different systems.



NTFS: Forensic Aspects



- NTFS file system is commonly used in digital forensics investigations to analyze and recover data from storage devices.
- NTFS provides a rich set of metadata, such as timestamps, permissions, and file attributes, which can provide valuable information to investigators.



NTFS: Forensic Aspects



- The Master File Table (MFT) is a critical component of NTFS forensic analysis, as it contains information about each file and directory on the volume.
- The MFT can be examined to determine the layout of files and directories on the volume, as well as to recover deleted files and folders.



NTFS: Forensic Aspects



- The \$LogFile, which contains information about changes made to the file system, can also be a valuable source of forensic evidence.
- NTFS forensic analysis can reveal a wealth of information about a system, including the operating system version, user accounts and passwords, application usage, and web browsing history.



NTFS: Forensic Aspects



- However, NTFS forensic analysis can also be challenging due to the complex structure of the file system and the potential for data hiding and obfuscation.
- Tools and techniques used in NTFS forensic analysis include disk imaging, file carving, keyword searching, timeline analysis, and metadata analysis.



NTFS: Forensic Aspects



- NTFS forensic analysis can be used in a variety of investigations, such as intellectual property theft, financial fraud, cybercrime, and child exploitation.



NTFS: Header Analysis



- The NTFS header is the first sector (sector 0) of the NTFS volume and contains important information about the file system, such as the number of sectors per cluster, the size of the MFT, and the location of the MFT mirror.
- NTFS header analysis is a crucial step in NTFS forensic analysis, as it provides information about the layout of the file system and can help identify anomalies or evidence of tampering.



NTFS: Header Analysis



- The NTFS header can be examined using various forensic tools, such as FTK Imager, EnCase, and dd.
- Important information that can be gleaned from the NTFS header includes the cluster size, which can affect the efficiency of file carving and recovery, and the location of the MFT mirror, which can be used to recover the MFT if it becomes corrupted.



NTFS: Header Analysis



- Anomalies in the NTFS header, such as incorrect values or signatures, can indicate potential tampering or damage to the file system.
- The NTFS header also contains a backup copy of the boot sector, which can be used to recover the file system if the primary boot sector becomes corrupted.



NTFS: Header Analysis



- In addition to the NTFS header, other important structures in NTFS forensic analysis include the MFT, \$LogFile, and \$Bitmap.
- Overall, NTFS header analysis is a critical component of NTFS forensic analysis and can provide valuable insights into the organization and integrity of the file system.



NTFS: Header Analysis



- In addition to the NTFS header, other important structures in NTFS forensic analysis include the MFT, \$LogFile, and \$Bitmap.
- Overall, NTFS header analysis is a critical component of NTFS forensic analysis and can provide valuable insights into the organization and integrity of the file system.



NTFS: MFT Analysis



- The MFT contains metadata for all files and directories on an NTFS volume, including information such as filename, creation time, modified time, access time, file size, permissions, and more.
- The MFT also contains information on deleted files, which can be useful in forensic investigations.



NTFS: MFT Analysis



- To perform forensic analysis of the MFT, forensic tools typically read the MFT entries and extract the relevant metadata.
- Some of the information that can be extracted from the MFT includes:
 - File and directory names: The MFT stores the names of all files and directories on the volume, including deleted ones.
 - File attributes: The MFT contains information on the attributes of each file, such as whether it is read-only, hidden, compressed, or encrypted.



NTFS: MFT Analysis



- Some of the information that can be extracted from the MFT includes:
 - Time stamps: The MFT stores the creation time, modified time, and access time for each file.
 - File size: The MFT contains the size of each file.
 - File location: The MFT stores the starting cluster of each file on the volume.



NTFS: MFT Analysis



- Forensic analysis of the MFT can be used to recover deleted files, track user activity, and identify potentially malicious files.
- It can also be useful in identifying system configuration and tracking changes to the file system over time.



NTFS: MFT Analysis



- However, it's important to note that the MFT can be subject to tampering or modification, so forensic analysts must take appropriate measures to preserve the integrity of the data.



NTFS: \$LogFile Analysis



- The \$LogFile is a system file in NTFS that records changes made to the file system, such as file creations, modifications, and deletions.
- The \$LogFile is an important source of forensic evidence in NTFS analysis, as it can provide a timeline of activity on the system and help identify potential indicators of compromise.



NTFS: \$LogFile Analysis



- The \$LogFile is a circular log, meaning that when it reaches a certain size, it overwrites the oldest entries with new ones.
- \$LogFile forensic analysis involves extracting and analyzing the contents of the log file to determine the history of activity on the file system.



NTFS: \$LogFile Analysis



- Various forensic tools, such as Encase and FTK Imager, can be used to extract and analyze the \$LogFile.
- \$LogFile forensic analysis can reveal valuable information, such as the names and paths of files that were created or modified, the timestamps of these events, and the identities of the users who performed them.



NTFS: \$LogFile Analysis



- \$LogFile forensic analysis can also help identify potential signs of malicious activity, such as files being renamed or moved to unusual locations, or the creation of new user accounts.
- However, \$LogFile forensic analysis can be challenging due to the large volume of data contained in the file and the complex structure of the log records.



NTFS: \$LogFile Analysis



- Various techniques can be used to improve the efficiency of \$LogFile forensic analysis, such as filtering the log records based on specific criteria or using timeline analysis to identify patterns of activity over time.



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- Various techniques can be used to improve the efficiency of \$LogFile forensic analysis, such as filtering the log records based on specific criteria or using timeline analysis to identify patterns of activity over time.



NTFS: \$Bitmap Analysis



- The \$Bitmap is a system file in NTFS that tracks the allocation status of clusters on the volume.
- Each bit in the \$Bitmap corresponds to a cluster on the volume, with a value of 1 indicating that the cluster is allocated and a value of 0 indicating that it is free.



NTFS: \$Bitmap Analysis



- The \$Bitmap is an important component of NTFS forensic analysis, as it can provide information about the allocation status of clusters on the volume, which can be used to recover deleted files and to identify potential areas of unallocated space where data may be hidden.
- The \$Bitmap is typically located in a fixed position on the volume and is relatively small in size compared to other system files, making it easier to extract and analyze.



NTFS: \$Bitmap Analysis



- \$Bitmap analysis can help identify potential areas of unallocated space on the volume where data may be hidden, such as in slack space or unallocated clusters.
- However, it is important to note that each cluster can only be allocated to one file or directory in NTFS, and a single file or directory cannot span multiple clusters that are already allocated to other files or directories.



NTFS: \$Bitmap Analysis



- When NTFS allocates a cluster to a file, it marks the corresponding bit in the \$Bitmap file as "1," indicating that the cluster is allocated to that file. If another file tries to allocate the same cluster, NTFS will not allow it, and the operation will fail.
- \$Bitmap analysis can also be used to recover deleted files and to identify potential signs of tampering or data hiding, such as the presence of allocated clusters in areas that should be unallocated.



NTFS: \$Bitmap Analysis



- However, \$Bitmap analysis can be complicated by the fact that clusters may be allocated for alternate data streams or for system files, which are not reflected in the \$Bitmap. Additionally, file compression or encryption can also affect the allocation of clusters on the volume, making it more difficult to determine which clusters are allocated to which files.



NTFS: ADS



- Alternate data streams (ADS) are a feature of NTFS that allows files to have multiple data streams. These streams can contain additional data, such as metadata or user-defined attributes, that is associated with the file. When an alternate data stream is created, it may consume one or more clusters, which are not reflected in the \$Bitmap.



NTFS: ADS



- In NTFS, a file can have multiple named data streams, which are essentially separate pieces of data that are associated with a file.
- These data streams can be used for various purposes such as storing metadata, indexing information, or even hiding data.



NTFS: ADS



- For example, a file can have an ADS that contains indexing information, allowing faster searching and retrieval of the file.
- ADS can also be used to store thumbnails of images, allowing faster previewing of images without opening the actual file.



NTFS: ADS



- The use of ADS in NTFS provides greater flexibility and functionality for storing and organizing data, making it a valuable feature for many applications and systems.



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FIXUP VALUES



- NTFS incorporates fixup values into data structures that are over one sector in length.
- With fixup values, the last two bytes of each sector in large data structures are replaced with a signature value when the data structure is written to disk.
- The signature is later used to verify the integrity of the data by verifying that all sectors have the same signature.
- Fixups are used only in data structures and not in sectors that contain file content.



FIXUP VALUES



- The data structures that use fixups have header fields that identify the current 16-bit signature value and an array that contains the original values.
- When the data structure is written to disk, the signature value is incremented by one, the last two bytes of each sector are copied to the array, and the signature value is written to the last two bytes of each sector.



FIXUP VALUES



- When reading the data structure, the OS should verify that the last two bytes of each sector are equal to the signature value, and the original values are then replaced from the array.
- Fixups are used to detect damaged sectors and corrupt data structures.
- If only one sector of a multi-sector data structure was written, the fixup will be different from the signature, and the OS will know that the data are corrupt.



FIXUP VALUES



- Each MFT entry uses fixup values, so the on-disk version of the data structure has the last two bytes of each sector replaced by a fixup value.



MFT Entry Address



- Each MFT entry uses fixup values, so the on-disk version of the data structure has the last two bytes of each sector replaced by a fixup value.



NTFS: Standard MFT Entry



Entry	File Name	Description
0	\$MFT	The entry for the MFT itself.
1	\$MFTMirr	Contains a backup of the first entries in the MFT.
2	\$LogFile	Contains the journal that records the metadata transactions.
3	\$Volume	Contains the volume information such as the label, identifier, and version.
4	\$AttrDef	Contains the attribute information, such as the identifier values, name, and sizes.
5	.	Contains the allocation status of each cluster in the file system. Contains the root directory of the file system.
6	\$Bitmap	



NTFS: Standard MFT Entry



Entry	File Name	Description
7	\$Boot	Contains the boot sector and boot code for the file system.
8	\$BadClus	Contains the clusters that have bad sectors.
9	\$Secure	Contains information about the security and access control for the files (Windows 2000 and XP version only).
10	\$Upcase	Contains the uppercase version of every Unicode character.
11	\$Extend	A directory that contains files for optional extensions. Microsoft does not typically place the files in this directory into the reserved MFT entries.



MFT Entry: Data structure



Byte Range	Description	Essential
0-3	Signature ("FILE")	No
4-5	Offset to fixup array.	Yes
6-7	Number of entries in fixup array	Yes
8-15	\$LogFile Sequence Number (LSN)	No
16-17	Sequence value	No
18-19	Link count	No
20-21	Offset to first attribute	Yes



MFT Entry: Data structure



Byte Range	Description	Essential
22-23	Flags (in-use and directory)	Yes
24-27	Used size of MFT entry	Yes
28-31	Allocated size of MFT entry	Yes
32-39	File reference to base record	No
40-41	Next attribute id	No
42-1023	Attributes and fixup values	Yes



NTFS: Attribute Headers



- The attribute header identifies the type of attribute, its size, and its name.
- It also has flags to identify if the value is compressed or encrypted.
- The attribute type is a numerical identifier based on the data type.
- An MFT entry can have multiple attributes of the same type.



NTFS: Attribute Headers



- Some attributes can be assigned a name and stored in UTF-16 Unicode in the attribute header.
- An attribute also has an identifier value assigned to it that is unique to that MFT entry.
- If an entry has more than one attribute of the same type, this identifier can be used to differentiate between them.



NTFS: Attribute Contents



- The content of the attribute can have any format and any size.
- NTFS provides two locations where attribute content can be stored.
- A resident attribute stores its content in the MFT entry with the attribute header.
- This works for only small attributes.



NTFS: Attribute Contents



- A non-resident attribute stores its content in an external cluster in the file system.
- The header of the attribute identifies if the attribute is resident or non-resident.
- If an attribute is resident, the content will immediately follow the header.



NTFS: Attribute Contents



- If the attribute is non-resident, the header will give the cluster addresses.
- Non-resident attributes are stored in cluster runs, which are consecutive clusters, and the run is documented using the starting cluster address and run length.



NTFS: Attribute Contents



- NTFS uses VCN-to-LCN mappings to describe the non-resident attribute runs.
- Logical Cluster Number (LCN) is the same as the logical file system address.
- Virtual Cluster Number (VCN) is the same as a logical file address.



NTFS: Standard Attribute Types



- A number is defined for each type of attribute, and Microsoft sorts the attributes in an entry using this number.
- The standard attributes have a default type value assigned to them.
- It can be redefined in the \$AttrDef file system metadata file.
- Each attribute type has a name, and it has all capital letters and starts with “\$.”



NTFS: Standard Attribute Types



- Not all these attribute types and identifiers will exist for every file.



NTFS: Standard Attribute Types



Entry	File Name	Description
16	\$STANDARD_INFORMATION	General information, such as flags; the last accessed, written, and created times; and the owner and security ID.
32	\$ATTRIBUTE_LIST	List where other attributes for the file can be found.
48	\$FILE_NAME	File name, in Unicode, and the last accessed, written, and created times.
64	\$VOLUME_VERSION	Volume information. Exists only in version 1.2 (Windows NT).
64	\$OBJECT_ID	A 16-byte unique identifier for the file or directory. Exists only in versions 3.0+ and after (Windows 2000+).
80	\$SECURITY_DESCRIPTOR	The access control and security properties of the file.
96	\$VOLUME_NAME	Volume name.



NTFS: Standard Attribute Types



Entry	File Name	Description
112	\$VOLUME_INFORMATION	File system version and other flags.
128	\$DATA	File contents.
144	\$INDEX_ROOT	Root node of an index tree.
160	\$INDEX_ALLOCATION	Nodes of an index tree rooted in \$INDEX_ROOT attribute.
176	\$BITMAP	A bitmap for the \$MFT file and for indexes.
192	\$SYMBOLIC_LINK	Exists only in version 1.2 (Windows NT).
192	\$REPARSE_POINT	Contains data about a reparse point, which is used as a soft link in version 3.0+ (Windows 2000+). Soft link information.



NTFS: Standard Attribute Types



Entry	File Name	Description
208	\$EA_INFORMATION	Used for backward compatibility with OS/2 applications (HPFS).
224	\$EA	Used for backward compatibility with OS/2 applications (HPFS).
256	\$LOGGED_UTILITY_STREAM	Contains keys and information about encrypted attributes in version 3.0+ (Windows 2000+).



Attribute: Data structure (0-16 Bytes)



Byte Range	Description	Essential
0-3	Attribute type identifier	No
4-7	Length of attribute	Yes
8-8	Non-resident flag	Yes
9-9	Length of name	No
10-11	Offset to name	No
12-13	Flags	No
14-15	Attribute identifier	Yes



Resident Attribute: Data structure



Byte Range	Description	Essential
0-15	General Header	Yes
16-19	Size of Content	Yes
20-21	Offset to content	Yes



Non-Resident Attribute: Data structure



Byte Range	Description	Essential
0-15	General Header	Yes
16-19	Starting VCN of the runlist	Yes
20-21	Ending VCN of the runlist	Yes
32-22	Offset to the runlist	Yes
34-35	Compression unit size	Yes
36-39	Unused	No
40-47	Allocated size of attribute content	No
48-55	Actual size of attribute content	Yes
56-63	Initialized size of attribute content	No



MFT Entry: Data structure



Flag values for the directory entry attributes field.		
Flag Value (in bits)	Description	Essential
0000 0000 0000 0001 (0 x 00 01)	Read only	No
0000 0000 0000 0010 (0 x 00 02)	Hidden file	No
0000 0000 0000 0100 (0 x 00 04)	System file	No
0000 0000 0010 0000 (0x 00 20)	Archive	No
0000 0000 0100 0000 (0 x 00 40)	Device	No
0000 0000 1000 0000 (0 x 00 80)	Normal	No
0000 0001 0000 0000 (0 x 01 00)	Temporary	No



MFT Entry: Data structure



Flag values for the directory entry attributes field.		
Flag Value (in bits)	Description	Essential
0010 0000 0000 0000 (0 x 20 00)	Sparse file	No
0100 0000 0000 0000 (0 x 40 00)	Reparse point	No
0000 1000 0000 0000 (0 x 08 00)	Compressed	No
0001 0000 0000 0000 (0 x 10 00)	Offline	No
0010 0000 0000 0000 (0 x 20 00)	Content is not being indexed for faster searches	No
0100 0000 0000 0000 (0x 40 00)	Encrypted	No



NTFS: MFT Entry Analysis

0bc00	46 49 4C 45 80	00 03 00-0B 85 00 02 00 00 00 00	FILE
0bc10	00 00 01 00 38	00 00 00-78 01 00 00 00 04 00 00	----
0bc20	00 00 00 00 00	00 00 00-04 00 00 00 2F 00 00 00	-----/---
0bc30	04 00 00 00 00	00 00 00-10 00 00 00 60 00 00 00	-----
0bc40	00 00 00 00 00	00 00 00-48 00 00 00 18 00 00 00	-----H----
0bc50	00 32 B3 6D 5E F9 D5 01-00 32 B3 6D 5E F9 D5 01		2'm^u0 2'm^u0
0bc60	47 66 69 A8 92 9E D9 01-C6 DC A8 92 9E D9 01		Gf1" -Ü .EÜ* -Ü
0bc70	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00		-----
0bc80	00 00 00 00 08 01 00 00-00 00 00 00 00 00 00		-----
0bc90	00 00 00 00 00 00 00 00-30 00 00 00 88 00 00 00		-----0-----
0bca0	00 00 00 00 00 00 02 00-70 00 00 00 18 00 01 00		-----p-----
0bcb0	05 00 00 00 00 00 05 00-F5 62 57 A8 92 9E D9 01		-----8bW" -Ü
0bcc0	F5 62 57 A8 92 9E D9 01-F5 62 57 A8 92 9E D9 01		8bW" -Ü 8bW" -Ü
0bcd0	F5 62 57 A8 92 9E D9 01-00 00 00 00 00 00 00		8bW" -Ü
0bce0	00 00 00 00 00 00 00 00-20 00 00 00 00 00 00		-----
0bcf0	17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00		--I-M-G_-2-0-2-
0bd00	30 00 30 00 33 00 31 00-30 00 5F 00 31 00 32 00		0-0-3-1-0_-1-2-
0bd10	31 00 31 00 33 00 33 00-2E 00 6A 00 70 00 67 00		1-1-3-3-_-j-p-g-
0bd20	80 00 00 00 50 00 00 00-01 00 00 00 00 03 00		-----P-----
0bd30	00 00 00 00 00 00 00 00-BD 03 00 00 00 00 00		-----s-----
0bd40	40 00 00 00 00 00 00 00-00 E0 3B 00 00 00 00		@-----à;-----
0bd50	94 D5 3B 00 00 00 00 00-94 D5 3B 00 00 00 00		-0;-----0;-----
0bd60	21 16 BA 53 22 A8 03 48-09 00 00 00 00 00 00		!-°S" -H-----
0bd70	FF FF FF FF 82 79 47 11-00 00 00 00 00 00 00		yyyy-yG-----
0bd80	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00		-----
0bd90	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00		-----
0bda0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00		-----
0bdb0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00		-----
0bdc0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00		-----
0bdd0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00		-----
0bde0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00		-----
0bdf0	00 00 00 00 00 00 00 00-00 00 00 00 00 04 00		-----

Signature of MFT Entry 0x46494c45



NTFS: MFT Entry Analysis

0c000	46 49 4C 45 30 00 03 00-7A 84 00 02 00 00 00 00	FILE0 -z-----
0c010	02 00 01 00 38 00 00 00-78 01 00 00 00 04 00 00	---8---x-----
0c020	00 00 00 00 00 00 00 00-04 00 00 00 30 00 00 00	-----0-----
0c030	03 00 00 00 00 00 00 00-10 00 00 00 60 00 00 00	-----^-----
0c040	00 00 00 00 00 00 00 00-48 00 00 00 18 00 00 00	-----H-----
0c050	00 32 B3 6D 5E F9 D5 01-00 32 B3 6D 5E F9 D5 01	·2³m^ûÖ·2³m^ûÖ·
0c060	F9 4D 82 A8 92 9E D9 01-00 32 B3 6D 5E F9 D5 01	ûM····Û·2³m^ûÖ·
0c070	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0c080	00 00 00 00 08 01 00 00-00 00 00 00 00 00 00	-----
0c090	00 00 00 00 00 00 00 00-30 00 00 00 88 00 00 00	-----0-----
0c0a0	00 00 00 00 00 00 02 00-70 00 00 00 18 00 01 00	-----p-----
0c0b0	05 00 00 00 00 00 05 00-B1 0D 6D A8 92 9E D9 01	-----±·m···û·
0c0c0	B1 0D 6D A8 92 9E D9 01-B1 0D 6D A8 92 9E D9 01	±·m···Û·±·m···Û·
0c0d0	B1 0D 6D A8 92 9E D9 01-00 00 00 00 00 00 00	±·m···Û·-----
0c0e0	00 00 00 00 00 00 00 00-20 00 00 00 00 00 00	-----
0c0f0	17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00	·I·M·G··2·0·2·
0c100	30 00 30 00 33 00 31 00-30 00 5F 00 31 00 33 00	0·0·3·1·0··1·3·
0c110	33 00 33 00 31 00 35 00-2E 00 6A 00 70 00 67 00	3·3·1·5··j·p·g·
0c120	80 00 00 00 50 00 00 00-01 00 00 00 00 00 03 00	-----P-----
0c130	00 00 00 00 00 00 00 00-24 05 00 00 00 00 00 00	-----\$-----
0c140	40 00 00 00 00 00 00 00-00 50 52 00 00 00 00 00	@·-----PR-----
0c150	0C 49 52 00 00 00 00 00-0C 49 52 00 00 00 00 00	·IR·-----IR-----
0c160	21 1E E2 58 22 07 05 CA-07 00 00 00 00 00 00 00	!·âX"··Ê·-----
0c170	FF FF FF FF 82 79 47 11-00 00 00 00 00 00 00 00	ÿÿÿÿ·yG·-----
0c180	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0c190	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0cla0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0clb0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0clc0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0cld0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0cle0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0clf0	00 00 00 00 00 00 00 00-00 00 00 00 00 03 00	-----

Offset to fixup array



NTFS: MFT Entry Analysis

0c000	46 49 4C 45 30 00 03 00 7A 84 00 02 00 00 00 00	FILE0 z
0c010	02 00 01 00 38 00 00 00 7A 01 00 00 00 04 00 00	---8---x---
0c020	00 00 00 00 00 00 00 00 00 04 00 00 00 30 00 00	-----0---
0c030	03 00 00 00 00 00 00 00 00 10 00 00 00 60 00 00	-----`---
0c040	00 00 00 00 00 00 00 00 00 48 00 00 00 18 00 00	-----H---
0c050	00 32 B3 6D 5E F9 D5 01 00 32 B3 6D 5E F9 D5 01	·2³m^uÖ·2³m^uÖ·
0c060	F9 4D 82 A8 92 9E D9 01 00 32 B3 6D 5E F9 D5 01	uM···Ü·2³m^uÖ·
0c070	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	-----
0c080	00 00 00 00 08 01 00 00 00 00 00 00 00 00 00 00	-----
0c090	00 00 00 00 00 00 00 00 00 30 00 00 00 88 00 00	-----0-----
0c0a0	00 00 00 00 00 00 02 00 70 00 00 00 18 00 01 00	-----p-----
0c0b0	05 00 00 00 00 00 05 00 B1 0D 6D A8 92 9E D9 01	-----±m··Ü·
0c0c0	B1 0D 6D A8 92 9E D9 01 B1 0D 6D A8 92 9E D9 01	±m··Ü·±m·Ü·
0c0d0	B1 0D 6D A8 92 9E D9 01 00 00 00 00 00 00 00 00	±m··Ü·-----
0c0e0	00 00 00 00 00 00 00 00 00 20 00 00 00 00 00 00	-----
0c0f0	17 00 49 00 4D 00 47 00 5F 00 32 00 30 00 32 00	·I·M·G·_·2·0·2·
0c100	30 00 30 00 33 00 31 00 30 00 5F 00 31 00 33 00	0·0·3·1·0·_·1·3·
0c110	33 00 33 00 31 00 35 00 2E 00 6A 00 70 00 67 00	3·3·1·5·_·j·p·g·
0c120	80 00 00 00 50 00 00 00 01 00 00 00 00 00 03 00	---P-----
0c130	00 00 00 00 00 00 00 00 00 24 05 00 00 00 00 00	-----\$-----
0c140	40 00 00 00 00 00 00 00 00 00 50 52 00 00 00 00	@-----PR-----
0c150	0C 49 52 00 00 00 00 00 00 0C 49 52 00 00 00 00	·IR-----IR-----
0c160	21 1E E2 58 22 07 05 CA 07 00 00 00 00 00 00 00	!·âX"··Ê-----
0c170	FF FF FF FF 82 79 47 11 00 00 00 00 00 00 00 00	ÿÿÿÿ·yG-----
0c180	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	-----
0c190	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	-----
0cla0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	-----
0clb0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	-----
0clc0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	-----
0cld0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	-----
0cle0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	-----
0clf0	00 00 00 00 00 00 00 00 00 00 00 00 00 00 03 00	-----

Number of entries in
fixup array



NTFS: MFT Entry Analysis

0c000	46 49 4C 45 30 00 03 00-7A 84 00 02 00 00 00 00	FILE0 -z
0c010	02 00 01 00 38 00 00 00-78 01 00 00 00 04 00 00	...8...x.....
0c020	00 00 00 00 00 00 00 00-04 00 00 00 30 00 00 00	0...
0c030	03 00 00 00 00 00 00 00-10 00 00 00 60 00 00 00	
0c040	00 00 00 00 00 00 00 00-48 00 00 00 18 00 00 00	H.....
0c050	00 32 B3 6D 5E F9 D5 01-00 32 B3 6D 5E F9 D5 01	·2'm^ûÖ·2'm^ûÖ·
0c060	F9 4D 82 A8 92 9E D9 01-00 32 B3 6D 5E F9 D5 01	ûM···û·2'm^ûÖ·
0c070	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c080	00 00 00 00 08 01 00 00-00 00 00 00 00 00 00	
0c090	00 00 00 00 00 00 00 00-30 00 00 00 88 00 00 00	0.....
0c0a0	00 00 00 00 00 00 02 00-70 00 00 00 18 00 01 00	p.....
0c0b0	05 00 00 00 00 00 05 00-B1 0D 6D A8 92 9E D9 01	±·m···Û·
0c0c0	B1 0D 6D A8 92 9E D9 01-B1 0D 6D A8 92 9E D9 01	±·m···Û·±·m···Û·
0c0d0	B1 0D 6D A8 92 9E D9 01-00 00 00 00 00 00 00	±·m···Û·.....
0c0e0	00 00 00 00 00 00 00 00-20 00 00 00 00 00 00	
0c0f0	17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00	·I·M·G·_·2·0·2·
0c100	30 00 30 00 33 00 31 00-30 00 5F 00 31 00 33 00	0·0·3·1·0·_·1·3·
0c110	33 00 33 00 31 00 35 00-2E 00 6A 00 70 00 67 00	3·3·1·5·_·j·p·g·
0c120	80 00 00 00 50 00 00 00-01 00 00 00 00 00 03 00	...P.....
0c130	00 00 00 00 00 00 00 00-24 05 00 00 00 00 00 00	\$.....
0c140	40 00 00 00 00 00 00 00-00 50 52 00 00 00 00 00	@.....PR.....
0c150	0C 49 52 00 00 00 00 00-0C 49 52 00 00 00 00 00	·IR.....IR.....
0c160	21 1E E2 58 22 07 05 CA-07 00 00 00 00 00 00 00	!·âX"··Ê.....
0c170	FF FF FF FF 82 79 47 11-00 00 00 00 00 00 00 00	ÿÿÿÿ·yG.....
0c180	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c190	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c1a0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c1b0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c1c0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c1d0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c1e0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c1f0	00 00 00 00 00 00 00 00-00 00 00 00 00 03 00	

\$LogFile Sequence
Number (LSN)



NTFS: MFT Entry Analysis

0c000	46 49 4C 45 30 00 03 00-7A 84 00 02 00 00 00 00	FILE0 ...z.....
0c010	02 00 01 00 38 00 00 00-78 01 00 00 00 04 00 00	...8...x.....
0c020	00 00 00 00 00 00 00 00-04 00 00 00 30 00 00 00	0.....
0c030	03 00 00 00 00 00 00 00-10 00 00 00 60 00 00 00	H.....
0c040	00 00 00 00 00 00 00 00-48 00 00 00 18 00 00 00	2³m^uÖ·-2³m^uÖ·
0c050	00 32 B3 6D 5E F9 D5 01-00 32 B3 6D 5E F9 D5 01	uM·-·Ü·-2³m^uÖ·
0c060	F9 4D 82 A8 92 9E D9 01-00 32 B3 6D 5E F9 D5 01	0.....
0c070	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	p.....
0c080	00 00 00 00 00 08 01 00 00-00 00 00 00 00 00 00	±·m·-·Ü·
0c090	00 00 00 00 00 00 00 00-30 00 00 00 88 00 00 00	±·m·-·Ü·±·m·-·Ü·
0c0a0	00 00 00 00 00 00 02 00-70 00 00 00 18 00 01 00	±·m·-·Ü·
0c0b0	05 00 00 00 00 00 05 00-B1 0D 6D A8 92 9E D9 01	I·M·G·-2·0·2·
0c0c0	B1 0D 6D A8 92 9E D9 01-B1 0D 6D A8 92 9E D9 01	0·0·3·1·0·-1·3·
0c0d0	B1 0D 6D A8 92 9E D9 01-00 00 00 00 00 00 00 00	3·3·1·5·-·j·p·g·
0c0e0	00 00 00 00 00 00 00 00-20 00 00 00 00 00 00 00	P.....
0c0f0	17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00	\$.....
0c100	30 00 30 00 33 00 31 00-30 00 5F 00 31 00 33 00	@·.....PR.....
0c110	33 00 33 00 31 00 35 00-2E 00 6A 00 70 00 67 00	·IR.....IR.....
0c120	80 00 00 00 50 00 00 00-01 00 00 00 00 00 03 00	!·âX"·-Ê.....
0c130	00 00 00 00 00 00 00 00-24 05 00 00 00 00 00 00	ÿÿÿÿ·yG.....
0c140	40 00 00 00 00 00 00 00-00 50 52 00 00 00 00 00	
0c150	0C 49 52 00 00 00 00 00-0C 49 52 00 00 00 00 00	
0c160	21 1E E2 58 22 07 05 CA-07 00 00 00 00 00 00 00	
0c170	FF FF FF FF 82 79 47 11-00 00 00 00 00 00 00 00	
0c180	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	
0c190	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	
0cla0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	
0clb0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	
0clc0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	
0cld0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	
0cle0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	
0clf0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 03 00	

Sequence value



NTFS: MFT Entry Analysis

0c000	46 49 4C 45 30 00 03 00-7A 84 00 02 00 00 00 00	FILE0 ---z-----
0c010	02 00 01 00 38 00 00 00-78 01 00 00 00 04 00 00	-8---x-----
0c020	00 00 00 00 00 00 00 00-04 00 00 00 30 00 00 00	-----0-----
0c030	03 00 00 00 00 00 00 00-10 00 00 00 60 00 00 00	-----`-----
0c040	00 00 00 00 00 00 00 00-48 00 00 00 18 00 00 00	-----H-----
0c050	00 32 B3 6D 5E F9 D5 01-00 32 B3 6D 5E F9 D5 01	-2³m^uÖ-2³m^uÖ-
0c060	F9 4D 82 A8 92 9E D9 01-00 32 B3 6D 5E F9 D5 01	uM- -Ü-2³m^uÖ-
0c070	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
0c080	00 00 00 00 08 01 00 00-00 00 00 00 00 00 00 00	-----
0c090	00 00 00 00 00 00 00 00-30 00 00 00 88 00 00 00	-----0-----
0c0a0	00 00 00 00 00 00 02 00-70 00 00 00 18 00 01 00	-----p-----
0c0b0	05 00 00 00 00 00 05 00-B1 0D 6D A8 92 9E D9 01	-----±·m- -Ü-
0c0c0	B1 0D 6D A8 92 9E D9 01-B1 0D 6D A8 92 9E D9 01	±·m- -Ü-±·m- -Ü-
0c0d0	B1 0D 6D A8 92 9E D9 01-00 00 00 00 00 00 00 00	±·m- -Ü-----
0c0e0	00 00 00 00 00 00 00 00-20 00 00 00 00 00 00 00	-----
0c0f0	17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00	-I·M·G- -2·0·2-
0c100	30 00 30 00 33 00 31 00-30 00 5F 00 31 00 33 00	0·0·3·1·0- -1·3-
0c110	33 00 33 00 31 00 35 00-2E 00 6A 00 70 00 67 00	3·3·1·5- -j·p·g-
0c120	80 00 00 00 50 00 00 00-01 00 00 00 00 00 03 00	-----P-----
0c130	00 00 00 00 00 00 00 00-24 05 00 00 00 00 00 00	-----\$-----
0c140	40 00 00 00 00 00 00 00-00 50 52 00 00 00 00 00	@-----PR-----
0c150	0C 49 52 00 00 00 00 00-0C 49 52 00 00 00 00 00	-IR-----IR-----
0c160	21 1E E2 58 22 07 05 CA-07 00 00 00 00 00 00 00	!·âX" -Ê-----
0c170	FF FF FF FF 82 79 47 11-00 00 00 00 00 00 00 00	ÿÿÿÿ·yG-----
0c180	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
0c190	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
0cla0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
0clb0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
0clc0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
0cld0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
0cle0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
0clf0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 03 00	-----

Link count



NTFS: MFT Entry Analysis

0c000	46 49 4C 45 30 00 03 00-7A 84 00 02 00 00 00 00	FILE0 ---z-----
0c010	02 00 01 00 38 00 00 00-78 01 00 00 00 04 00 00	---8---x-----
0c020	00 00 00 00 00 00 00 00-04 00 00 00 30 00 00 00	-----0-----
0c030	03 00 00 00 00 00 00 00-10 00 00 00 60 00 00 00	-----^-----
0c040	00 00 00 00 00 00 00 00-48 00 00 00 18 00 00 00	-----H-----
0c050	00 32 B3 6D 5E F9 D5 01-00 32 B3 6D 5E F9 D5 01	-2'm^uÖ-2'm^uÖ-
0c060	F9 4D 82 A8 92 9E D9 01-00 32 B3 6D 5E F9 D5 01	uM"-Û-2'm^uÖ-
0c070	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0c080	00 00 00 00 08 01 00 00-00 00 00 00 00 00 00	-----
0c090	00 00 00 00 00 00 00 00-30 00 00 00 88 00 00 00	-----0-----
0c0a0	00 00 00 00 00 00 02 00-70 00 00 00 18 00 01 00	-----p-----
0c0b0	05 00 00 00 00 00 05 00-B1 0D 6D A8 92 9E D9 01	-----m"-Û-
0c0c0	B1 0D 6D A8 92 9E D9 01-B1 0D 6D A8 92 9E D9 01	±m"-Û±m"-Û-
0c0d0	B1 0D 6D A8 92 9E D9 01-00 00 00 00 00 00 00	±m"-Û-----
0c0e0	00 00 00 00 00 00 00 00-20 00 00 00 00 00 00	-----
0c0f0	17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00	-I·M·G-2·0·2-
0c100	30 00 30 00 33 00 31 00-30 00 5F 00 31 00 33 00	0·0·3·1·0-1·3-
0c110	33 00 33 00 31 00 35 00-2E 00 6A 00 70 00 67 00	3·3·1·5-·j·p·g-
0c120	80 00 00 00 50 00 00 00-01 00 00 00 00 00 03 00	---P-----
0c130	00 00 00 00 00 00 00 00-24 05 00 00 00 00 00 00	-----\$-----
0c140	40 00 00 00 00 00 00 00-00 50 52 00 00 00 00 00	@-----PR-----
0c150	0C 49 52 00 00 00 00 00-0C 49 52 00 00 00 00 00	-IR-----IR-----
0c160	21 1E E2 58 22 07 05 CA-07 00 00 00 00 00 00 00	!·âX"-·Ê-----
0c170	FF FF FF FF 82 79 47 11-00 00 00 00 00 00 00 00	ÿÿÿÿ·yG-----
0c180	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0c190	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0cla0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0clb0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0clc0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0cld0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0cle0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0clf0	00 00 00 00 00 00 00 00-00 00 00 00 00 03 00	-----

Offset to first attribute



NTFS: MFT Entry Analysis

0c000	46 49 4C 45 30 00 03 00-FA 84 00 02 00 00 00 00	FILE0...z.....
0c010	02 00 01 00 38 00 00 00-78 01 00 00 00 04 00 00	...8...x.....
0c020	00 00 00 00 00 00 00 00-00 00 00 00 30 00 00 00	0.....
0c030	03 00 00 00 00 00 00 00-10 00 00 00 60 00 00 00	`.....
0c040	00 00 00 00 00 00 00 00-48 00 00 00 18 00 00 00	H.....
0c050	00 32 B3 6D 5E F9 D5 01-00 32 B3 6D 5E F9 D5 01	..2'm^uÖ..2'm^uÖ..
0c060	F9 4D 82 A8 92 9E D9 01-00 32 B3 6D 5E F9 D5 01	uM-..Û..2'm^uÖ..
0c070	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c080	00 00 00 00 08 01 00 00-00 00 00 00 00 00 00	
0c090	00 00 00 00 00 00 00 00-30 00 00 00 88 00 00 00	0.....
0c0a0	00 00 00 00 00 00 02 00-70 00 00 00 18 00 01 00	p.....
0c0b0	05 00 00 00 00 00 05 00-B1 0D 6D A8 92 9E D9 01	m..Û..
0c0c0	B1 0D 6D A8 92 9E D9 01-B1 0D 6D A8 92 9E D9 01	±·m"·Û·±·m"·Û·
0c0d0	B1 0D 6D A8 92 9E D9 01-00 00 00 00 00 00 00	±·m"·Û·
0c0e0	00 00 00 00 00 00 00 00-20 00 00 00 00 00 00	
0c0f0	17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00	..I·M·G_..2·0·2·
0c100	30 00 30 00 33 00 31 00-30 00 5F 00 31 00 33 00	0·0·3·1·0_·1·3·
0c110	33 00 33 00 31 00 35 00-2E 00 6A 00 70 00 67 00	3·3·1·5·.·j·p·g·
0c120	80 00 00 00 50 00 00 00-01 00 00 00 00 00 03 00	...P.....
0c130	00 00 00 00 00 00 00 00-24 05 00 00 00 00 00 00	\$.....
0c140	40 00 00 00 00 00 00 00-00 50 52 00 00 00 00 00	@.....PR.....
0c150	0C 49 52 00 00 00 00 00-0C 49 52 00 00 00 00 00	·IR.....IR.....
0c160	21 1E E2 58 22 07 05 CA-07 00 00 00 00 00 00 00	!·âx"·-Ê.....
0c170	FF FF FF FF 82 79 47 11-00 00 00 00 00 00 00 00	ÿÿÿÿ·yG.....
0c180	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c190	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c1a0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c1b0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c1c0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c1d0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c1e0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	
0c1f0	00 00 00 00 00 00 00 00-00 00 00 00 00 03 00	

Flags (in-use and directory)



NTFS: MFT Entry Analysis

0c000	46 49 4C 45 30 00 03 00	7A 84 00 02 00 00 00 00	FILE0	z
0c010	02 00 01 00 38 00 00 00	00 78 01 00 00 00 00 00	---	8 x
0c020	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	---	0
0c030	03 00 00 00 00 00 00 00	00 10 00 00 00 60 00 00	---	
0c040	00 00 00 00 00 00 00 00	00 48 00 00 00 18 00 00	---	H
0c050	00 32 B3 6D 5E F9 D5 01	00 32 B3 6D 5E F9 D5 01	2'm^uÖ	2'm^uÖ
0c060	F9 4D 82 A8 92 9E D9 01	00 32 B3 6D 5E F9 D5 01	uM	2'm^uÖ
0c070	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	---	
0c080	00 00 00 00 08 01 00 00	00 00 00 00 00 00 00 00	---	
0c090	00 00 00 00 00 00 00 00	00 30 00 00 00 88 00 00	---	0
0c0a0	00 00 00 00 00 00 02 00	00 70 00 00 00 18 00 01	---	
0c0b0	05 00 00 00 00 00 05 00	00 B1 0D 6D A8 92 9E D9 01	±·m	Ü
0c0c0	B1 0D 6D A8 92 9E D9 01	B1 0D 6D A8 92 9E D9 01	±·m	Ü ±·m
0c0d0	B1 0D 6D A8 92 9E D9 01	00 00 00 00 00 00 00 00	±·m	Ü
0c0e0	00 00 00 00 00 00 00 00	00 20 00 00 00 00 00 00	---	
0c0f0	17 00 49 00 4D 00 47 00	00 5F 00 32 00 30 00 32 00	I·M·G	2·0·2
0c100	30 00 30 00 33 00 31 00	00 30 00 5F 00 31 00 33 00	0·0·3·1·0	1·3
0c110	33 00 33 00 31 00 35 00	00 2E 00 6A 00 70 00 67 00	3·3·1·5	j·p·g
0c120	80 00 00 00 50 00 00 00	00 01 00 00 00 00 03 00	---	P
0c130	00 00 00 00 00 00 00 00	00 24 05 00 00 00 00 00	---	\$
0c140	40 00 00 00 00 00 00 00	00 00 50 52 00 00 00 00	@	PR
0c150	0C 49 52 00 00 00 00 00	00 0C 49 52 00 00 00 00	IR	IR
0c160	21 1E E2 58 22 07 05 CA	00 07 00 00 00 00 00 00	!·âX"	Ê
0c170	FF FF FF FF 82 79 47 11	00 00 00 00 00 00 00 00	ÿÿÿÿ	yG
0c180	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	---	
0c190	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	---	
0c1a0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	---	
0c1b0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	---	
0c1c0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	---	
0c1d0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	---	
0c1e0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	---	
0c1f0	00 00 00 00 00 00 00 00	00 00 00 00 00 00 03 00	---	

Used size of MFT entry
Value: 0 x 00 00 01 78
= 376 Bytes



NTFS: MFT Entry Analysis

0c000	46 49 4C 45 30 00 03 00-7A 84 00 02 00 00 00 00	FILE0 --z-----
0c010	02 00 01 00 38 00 00 00-78 01 00 00 04 00 00	--8--x-----
0c020	00 00 00 00 00 00 00 00-04 00 00 00 00 00 00	-----0----
0c030	03 00 00 00 00 00 00 00-10 00 00 00 60 00 00 00	-----H-----
0c040	00 00 00 00 00 00 00 00-48 00 00 00 18 00 00 00	-----2'm'uÖ-2'm'uÖ-
0c050	00 32 B3 6D 5E F9 D5 01-00 32 B3 6D 5E F9 D5 01	uM--'ü-2'm'uÖ-
0c060	F9 4D 82 A8 92 9E D9 01-00 32 B3 6D 5E F9 D5 01	-----0-----
0c070	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----p-----
0c080	00 00 00 00 00 08 01 00-00 00 00 00 00 00 00 00	-----±·m"-·ü-
0c090	00 00 00 00 00 00 00 00-30 00 00 00 88 00 00 00	-----±·m"-·ü±·m"-·ü-
0c0a0	00 00 00 00 00 00 02 00-70 00 00 00 18 00 01 00	-----±·m"-·ü-----
0c0b0	05 00 00 00 00 00 05 00-B1 0D 6D A8 92 9E D9 01	-----I·M·G-2·0·2-
0c0c0	B1 0D 6D A8 92 9E D9 01-B1 0D 6D A8 92 9E D9 01	0·0·3·1·0-1·3-
0c0d0	B1 0D 6D A8 92 9E D9 01-00 00 00 00 00 00 00 00	3·3·1·5-·j·p·g-
0c0e0	00 00 00 00 00 00 00 00-20 00 00 00 00 00 00 00	-----P-----
0c0f0	17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00	-----\$-----
0c100	30 00 30 00 33 00 31 00-30 00 5F 00 31 00 33 00	-----@-----PR-----
0c110	33 00 33 00 31 00 35 00-2E 00 6A 00 70 00 67 00	-----IR-----IR-----
0c120	80 00 00 00 50 00 00 00-01 00 00 00 00 00 03 00	-----!·âx"-·Ê-----
0c130	00 00 00 00 00 00 00 00-24 05 00 00 00 00 00 00	-----ÿÿÿÿ·yG-----
0c140	40 00 00 00 00 00 00 00-00 50 52 00 00 00 00 00	-----P-----
0c150	0C 49 52 00 00 00 00 00-0C 49 52 00 00 00 00 00	-----P-----
0c160	21 1E E2 58 22 07 05 CA-07 00 00 00 00 00 00 00	-----P-----
0c170	FF FF FF FF 82 79 47 11-00 00 00 00 00 00 00 00	-----P-----
0c180	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----P-----
0c190	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----P-----
0cla0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----P-----
0clb0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----P-----
0clc0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----P-----
0cld0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----P-----
0cle0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----P-----
0clf0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 03 00	-----P-----

Allocated size of MFT entry

Value = 0x00000400
= 1024 Bytes



NTFS: MFT Entry Analysis

0c000	46 49 4C 45 30 00 03 00-7A 84 00 02 00 00 00 00	FILE0 ---z-----
0c010	02 00 01 00 38 00 00 00-78 01 00 00 00 04 00 00	---8---x-----
0c020	00 00 00 00 00 00 00 00-04 00 00 00 30 00 00 00	-----0-----
0c030	03 00 00 00 00 00 00 00-10 00 00 00 60 00 00 00	-----H-----
0c040	00 00 00 00 00 00 00 00-48 00 00 00 18 00 00 00	-----2^m^uO--2^m^uO--
0c050	00 32 B3 6D 5E F9 D5 01-00 32 B3 6D 5E F9 D5 01	uM--"-Ü--2^m^uO--
0c060	F9 4D 82 A8 92 9E D9 01-00 32 B3 6D 5E F9 D5 01	-----0-----
0c070	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----p-----
0c080	00 00 00 00 08 01 00 00-00 00 00 00 00 00 00	---±·m"-·Ü·
0c090	00 00 00 00 00 00 00 00-30 00 00 00 88 00 00 00	±·m"-·Ü·±·m"-·Ü·
0c0a0	00 00 00 00 00 00 02 00-70 00 00 00 18 00 01 00	±·m"-·Ü·
0c0b0	05 00 00 00 00 00 05 00-B1 0D 6D A8 92 9E D9 01	---I·M·G·-2·0·2·
0c0c0	B1 0D 6D A8 92 9E D9 01-B1 0D 6D A8 92 9E D9 01	0·0·3·1·0·-1·3·
0c0d0	B1 0D 6D A8 92 9E D9 01-00 00 00 00 00 00 00	3·3·1·5·-·j·p·g·
0c0e0	00 00 00 00 00 00 00 00-20 00 00 00 00 00 00	---P-----
0c0f0	17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00	-----\$-----
0c100	30 00 30 00 33 00 31 00-30 00 5F 00 31 00 33 00	@-----PR-----
0c110	33 00 33 00 31 00 35 00-2E 00 6A 00 70 00 67 00	---IR-----IR-----
0c120	80 00 00 00 50 00 00 00-01 00 00 00 00 00 03 00	!·âX"-·Ê-----
0c130	00 00 00 00 00 00 00 00-24 05 00 00 00 00 00 00	ÿÿÿÿ·yG-----
0c140	40 00 00 00 00 00 00 00-00 50 52 00 00 00 00 00	-----
0c150	0C 49 52 00 00 00 00 00-0C 49 52 00 00 00 00 00	-----
0c160	21 1E E2 58 22 07 05 CA-07 00 00 00 00 00 00 00	-----
0c170	FF FF FF FF 82 79 47 11-00 00 00 00 00 00 00 00	-----
0c180	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0c190	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0cla0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0clb0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0clc0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0cld0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0cle0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00	-----
0clf0	00 00 00 00 00 00 00 00-00 00 00 00 00 03 00	-----

File reference to base
record



NTFS: MFT Entry Analysis

0c000	46 49 4C 45 30 00 03 00-7A 84 00 02 00 00 00 00	FILE0 ---z-----
0c010	02 00 01 00 38 00 00 00-78 01 00 00 00 04 00 00	---8---x-----
0c020	00 00 00 00 00 00 00 00-04 00 00 00 30 00 00 00	-----0-----
0c030	03 00 00 00 00 00 00 00-10 00 00 00 60 00 00 00	-----H-----
0c040	00 00 00 00 00 00 00 00-48 00 00 00 18 00 00 00	-----2'm^uÖ-2'm^uÖ-
0c050	00 32 B3 6D 5E F9 D5 01-00 32 B3 6D 5E F9 D5 01	uM---Ü-2'm^uÖ-
0c060	F9 4D 82 A8 92 9E D9 01-00 32 B3 6D 5E F9 D5 01	-----0-----
0c070	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----0-----
0c080	00 00 00 00 08 01 00 00-00 00 00 00 00 00 00 00	-----0-----
0c090	00 00 00 00 00 00 00 00-30 00 00 00 88 00 00 00	-----0-----
0c0a0	00 00 00 00 00 00 02 00-70 00 00 00 18 00 01 00	-----0-----
0c0b0	05 00 00 00 00 00 05 00-B1 0D 6D A8 92 9E D9 01	-----±·m---Ü-
0c0c0	B1 0D 6D A8 92 9E D9 01-B1 0D 6D A8 92 9E D9 01	±·m---Ü-±·m---Ü-
0c0d0	B1 0D 6D A8 92 9E D9 01-00 00 00 00 00 00 00 00	±·m---Ü-
0c0e0	00 00 00 00 00 00 00 00-20 00 00 00 00 00 00 00	-----0-----
0c0f0	17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00	·I·M·G·-2·0·2·
0c100	30 00 30 00 33 00 31 00-30 00 5F 00 31 00 33 00	0·0·3·1·0·-1·3·
0c110	33 00 33 00 31 00 35 00-2E 00 6A 00 70 00 67 00	3·3·1·5·-·j·p·g·
0c120	80 00 00 00 00 50 00 00-01 00 00 00 00 00 03 00	---P-----
0c130	00 00 00 00 00 00 00 00-24 05 00 00 00 00 00 00	-----\$-----
0c140	40 00 00 00 00 00 00 00-00 50 52 00 00 00 00 00	@-----PR-----
0c150	0C 49 52 00 00 00 00 00-0C 49 52 00 00 00 00 00	·IR-----IR-----
0c160	21 1E E2 58 22 07 05 CA-07 00 00 00 00 00 00 00	!·âX"-·Ê-----
0c170	FF FF FF FF 82 79 47 11-00 00 00 00 00 00 00 00	ÿÿÿÿ·yG-----
0c180	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----0-----
0c190	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----0-----
0cla0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----0-----
0clb0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----0-----
0clc0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----0-----
0cld0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----0-----
0cle0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----0-----
0clf0	00 00 00 00 00 00 00 00-00 00 00 00 00 03 00	-----0-----

Next attribute id

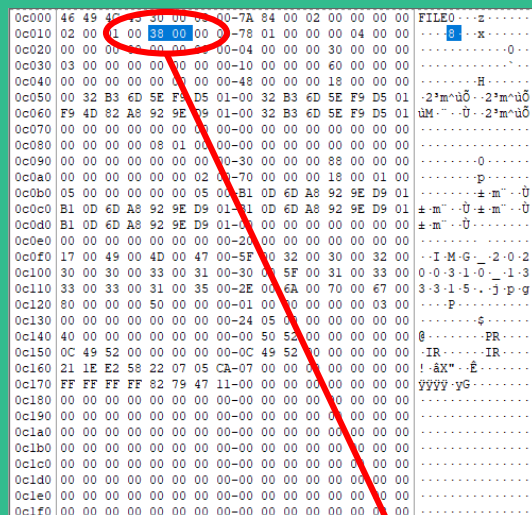


The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. At the top, the letters 'NFSU' are written in a serif font. The shield is divided into four quadrants: top-left (blue) with a white computer monitor icon, top-right (red) with a white fingerprint icon, bottom-left (red) with a white DNA double helix icon, and bottom-right (blue) with a white icon of a microscope. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in a curved, stylized font.

Attributes and fixup values



The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a yellow border and a central blue field divided into four quadrants. The top-left quadrant contains a white icon of a computer monitor with a magnifying glass. The top-right quadrant contains a white fingerprint icon. The bottom-left quadrant contains a white icon of a circular arrow, symbolizing a process or cycle. The bottom-right quadrant contains a white DNA double helix icon. Above the shield, the letters 'NFSU' are written in a bold, black, sans-serif font. Below the shield, the motto 'विद्यया अमृतं अश्नुते' is written in Devanagari script, which translates to 'Knowledge is the immortal nectar'.



38	00	00	00-78	01	00	00	00	04	00	00	...	8	...
00	00	00	00-04	00	00	00	30	00	00	00	...	...	...
00	00	00	00-10	00	00	00	60	00	00	00	...	...	...

sarang.rajvansh@nfsu.ac.in

Attribute Analysis

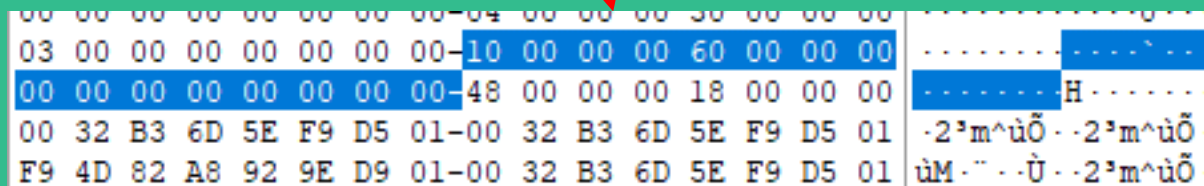
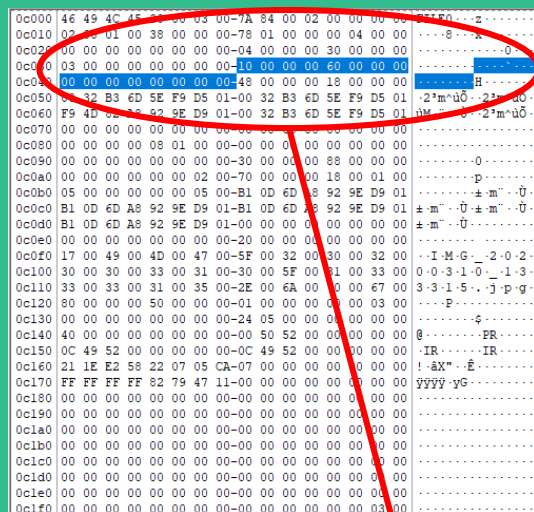


000	46 49 4C 45 30 00 03 00-7A 84 00 02 00 00 00 00	FILE0---z-----
010	02 00 01 00 38 00 00 00-78 01 00 00 00 04 00 00	---8---x-----
020	00 00 00 00 00 00 00 00-04 00 00 00 30 00 00 00	-----0-----
030	03 00 00 00 00 00 00 00-10 00 00 00 60 00 00 00	-----H-----
040	00 00 00 00 00 00 00 00-48 00 00 00 18 00 00 00	-----2'm^uÖ--2'm^uÖ-
050	00 32 B3 6D 5E F9 D5 01-00 32 B3 6D 5E F9 D5 01	üM---Ü--2'm^uÖ-
060	F9 4D 82 A8 92 9E D9 01-00 32 B3 6D 5E F9 D5 01	-----0-----
070	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----p-----
080	00 00 00 00 08 01 00 00-00 00 00 00 00 00 00 00	-----±-m"-Ü-
090	00 00 00 00 00 00 00 00-30 00 00 00 88 00 00 00	±-m"-Ü-±-m"-Ü-
0a0	00 00 00 00 00 00 02 00-70 00 00 00 18 00 01 00	±-m"-Ü-
0b0	05 00 00 00 00 00 05 00-B1 0D 6D A8 92 9E D9 01	-----I-M-G_-2-0-2-
0c0	B1 0D 6D A8 92 9E D9 01-B1 0D 6D A8 92 9E D9 01	0-0-3-1-0_-1-3-
0d0	B1 0D 6D A8 92 9E D9 01-00 00 00 00 00 00 00 00	3-3-1-5-.j-p-g-
0e0	00 00 00 00 00 00 00 00-20 00 00 00 00 00 00 00	-----P-----
0f0	17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00	-----\$-----
100	30 00 30 00 33 00 31 00-30 00 5F 00 31 00 33 00	@-----PR-----
110	33 00 33 00 31 00 35 00-2E 00 6A 00 70 00 67 00	-IR-----IR-----
120	80 00 00 00 50 00 00 00-01 00 00 00 00 00 03 00	!-âX"-Ê-----
130	00 00 00 00 00 00 00 00-24 05 00 00 00 00 00 00	ÿÿÿÿ-yG-----
140	40 00 00 00 00 00 00 00-00 50 52 00 00 00 00 00	-----
150	0C 49 52 00 00 00 00 00-0C 49 52 00 00 00 00 00	-----
160	21 1E E2 58 22 07 05 CA-07 00 00 00 00 00 00 00	-----
170	FF FF FF FF 82 79 47 11-00 00 00 00 00 00 00 00	-----
180	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
190	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
1a0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
1b0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
1c0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
1d0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----
1e0	00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00	-----

Cursor pos = 56

First attribute byte offset 56 (0x0038).

The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. It features a gold border and a gold background. At the top, the letters "NFSU" are written in gold. The shield is divided into four quadrants by a gold cross. The top-left quadrant is blue and contains a white icon of a computer monitor. The top-right quadrant is red and contains a white fingerprint icon. The bottom-left quadrant is red and contains a white icon of a microscope. The bottom-right quadrant is blue and contains a white DNA double helix icon. Below the shield, the motto "विद्यया अमृतं अश्नुते" is written in Devanagari script.



National Forensic Sciences University

The logo of the National Forensic Science University (NFSU) is a shield-shaped emblem. At the top, the letters "NFSU" are written in a serif font. The shield is divided into four quadrants: top-left (blue) with a white computer monitor icon, top-right (red) with a white fingerprint icon, bottom-left (red) with a white DNA double helix icon, and bottom-right (blue) with a white icon of a microscope. Below the shield, the Sanskrit motto "विद्यया अमृतं अश्नुते" (Vidyayā amṛtaṁ aśnute) is written in a curved, stylized font.



The screenshot displays a debugger window with assembly code on the left and CPU registers on the right. A red circle highlights the instruction at address 00000040: B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01. A red arrow points to the instruction at address 00000044: B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01. The right side of the window shows the CPU registers, including EAX, ECX, EDI, and ESI, with their current values and flags.

Address	Disassembly	Comment
00000040	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000044	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000048	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
0000004C	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000050	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000054	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000058	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
0000005C	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000060	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000064	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000068	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
0000006C	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000070	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000074	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000078	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
0000007C	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000080	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000084	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000088	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
0000008C	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000090	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000094	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
00000098	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
0000009C	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000A0	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000A4	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000A8	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000AC	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000B0	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000B4	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000B8	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000BC	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000C0	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000C4	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000C8	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000CC	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000D0	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000D4	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000D8	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000DC	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000E0	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000E4	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000E8	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000EC	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000F0	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000F4	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000F8	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	
000000FC	B3 6D 5E F9 05 01-00 32 B3 6D 5E F9 D5 01	

Registers:

Register	Value
EAX	00000000
ECX	00000000
EDX	00000000
EBX	00000000
ESI	00000000
EDI	00000000
EIP	00000040

Flags:

Flag	Value
CF	0
PF	0
AF	0
ZF	0
SF	0
OF	0

0-10	00	00	00	60	00	00	00	00	
0-48	00	00	00	18	00	00	00	00	H.....
1-00	32	B3	6D	5E	F9	D5	01	01	$\cdot 2^3 m^{\wedge} \tilde{u} \tilde{O} \cdot \cdot 2^3 m$

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Attribute Analysis

```
000 46 49 4C 45 30 00 03 00-7A 84 00 02 00 00 00 00 FILE0...z.....
010 02 00 01 00 38 00 00 00-78 01 00 00 00 04 00 00 ...8...x.....
020 00 00 00 00 00 00 00 00-04 00 00 00 00 00 00 ...0.....
030 03 00 00 00 00 00 00 00-10 00 00 60 00 00 00 .....0.....
040 00 00 00 00 00 00 00 00-48 00 00 18 00 00 00 .....H...
050 00 32 B3 6D 5E F9 D5 01-00 32 B3 6D 5E F9 D5 01 ...2'mu0...2'mu0...
060 F9 4D 82 A8 92 9E D9 01-00 32 B3 6D 5E F9 D5 01 uM...U...2'mu0...
070 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
080 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
090 00 00 00 00 00 00 00 00-30 00 00 88 00 00 00 .....0.....
0a0 00 00 00 00 00 00 00 02-00-70 00 00 18 00 01 00 .....p.....
0b0 05 00 00 00 00 00 05 00-B1 0D 6D A8 92 9E D9 01 ±m...U±m...U...
0c0 B1 0D 6D A8 92 9E D9 01-B1 0D 6D A8 92 9E D9 01 ±m...U±m...U...
0d0 B1 0D 6D A8 92 9E D9 01-00 00 00 00 00 00 00 ±m...U±m...U...
0e0 00 00 00 00 00 00 00 00-20 00 00 00 00 00 00 .....
0f0 17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00 ...I-M-G...2-0-2...
100 30 00 30 00 33 00 31 00-30 00 5F 00 31 00 33 00 0-0-3-1-0...1-3...
110 33 00 33 00 31 00 35 00-2E 00 6A 00 70 00 67 00 3-3-1-5...J-p-g...
120 80 00 00 00 50 00 00 00-01 00 00 00 00 00 00 ...P.....
130 00 00 00 00 00 00 00 00-24 05 00 00 00 00 00 .....$.....
140 40 00 00 00 00 00 00 00-00 50 52 00 00 00 00 0 8.....PR.....
150 0C 49 52 00 00 00 00 00-0C 49 52 00 00 00 00 00 -IR.....IR.....
160 21 1E E2 58 22 07 05 CA-07 00 00 00 00 00 00 !-AX*-E...
170 FF FF FF FF 82 79 47 11-00 00 00 00 00 00 00 yyy-yG...
180 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
190 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
1a0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
1b0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
1c0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
1d0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
1e0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 .....
1f0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 03 0
```

```
00 00 00 00 00 00
00 18 00 00 00 00 .....H...
0D 5F F9 D5 01 ...2'mu0...2'mu0...
```

Offset to content (20-21 Bytes)



The logo of the National Food Security University (NFSU) is a shield-shaped emblem. At the top, it features a golden crown-like structure. Below this, the acronym 'NFSU' is written in a bold, black, serif font. The shield is divided into four quadrants: the top-left is blue with a white computer monitor icon; the top-right is red with a white fingerprint icon; the bottom-left is red with a white circular arrow icon; and the bottom-right is blue with a white DNA double helix icon. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in a stylized, black, Devanagari script.

Value = 0 x 00 00 00 10 = \$STANDARD_INFORMATION

Value = 0 x 00 00 00 60 = 96 Bytes Length

Value = 0 x 00 = resident data

Value = 0 x 00 = 0 Bytes

Value = 0 x 00 00 = 0 Bytes

Value = 0 x 00 00 = 0 Bytes

Value = 0 x 00 00 = 0 Bytes

[illegible]

The logo of the National Food Security University (NFSU) is a shield-shaped emblem. At the top, it features a golden crown-like structure. Below this, the acronym 'NFSU' is written in a bold, black, serif font. The shield is divided into four quadrants: the top-left is blue with a white computer monitor icon; the top-right is red with a white fingerprint icon; the bottom-left is red with a white circular arrow icon; and the bottom-right is blue with a white DNA double helix icon. Below the shield, the Sanskrit motto 'विद्यया अमृतं अश्नुते' (Vidyayā amṛtaṁ aśnute) is written in a stylized, black, Devanagari script.

[illegible]

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Attribute Analysis: Content

```
0c000 46 49 4C 45 30 00 03 00-7A 04 00 02 00 00 00 00 00 FILE.....
0c010 02 00 01 00 38 00 00 00-78 01 00 00 00 04 00 00 00 .....8.....
0c020 00 00 00 00 00 00 00 00-04 00 00 00 00 30 00 00 00 .....0.....
0c030 03 00 00 00 00 00 00 00-10 00 00 00 00 60 00 00 00 .....0.....
0c040 00 00 00 00 00 00 00 00-48 00 00 00 00 18 00 00 00 .....8.....
0c050 00 32 B3 6D 5E F9 D5 01-00 32 B3 6D 5E F9 D5 01 2'm'u0-2'm'u0-
0c060 F9 4D 82 A8 92 9E D9 01-00 32 B3 6D 5E F9 D5 01 2'm'u0-2'm'u0-
0c070 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 .....0.....
0c080 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 .....0.....
0c090 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 .....0.....
0c0a0 00 00 00 00 00 00 00 00-02 00-70 00 00 00 18 00 01 00 .....p.....
0c0b0 05 00 00 00 00 00 00 05-00-B1 0D 6D A8 92 9E D9 01 .....±·m"-·Û·
0c0c0 B1 0D 6D A8 92 9E D9 01-B1 0D 6D A8 92 9E D9 01 ±·m"-·Û·±·m"-·Û·
0c0d0 B1 0D 6D A8 92 9E D9 01-00 00 00 00 00 00 00 00 00 ±·m"-·Û·.....
0c0e0 00 00 00 00 00 00 00 00-20 00 00 00 00 00 00 00 .....
0c0f0 17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00 ··I·M·G·_·2·0·2·
0c100 30 00 30 00 33 00 31 00-30 00 5F 00 31 00 33 00 0·0·3·1·0·_·1·3·
0c110 33 00 33 00 31 00 35 00-2E 00 6A 00 70 00 67 00 3·3·1·5·_·j·p·g·
```

\$FILENAME

Attribute Content: 136 Bytes

```
0c090 00 00 00 00 00 00 00 00-30 00 00 00 88 00 00 00 .....0.....
0c0a0 00 00 00 00 00 00 02 00-70 00 00 00 18 00 01 00 .....p.....
0c0b0 05 00 00 00 00 00 05 00-B1 0D 6D A8 92 9E D9 01 .....±·m"-·Û·
0c0c0 B1 0D 6D A8 92 9E D9 01-B1 0D 6D A8 92 9E D9 01 ±·m"-·Û·±·m"-·Û·
0c0d0 B1 0D 6D A8 92 9E D9 01-00 00 00 00 00 00 00 00 00 ±·m"-·Û·.....
0c0e0 00 00 00 00 00 00 00 00-20 00 00 00 00 00 00 00 .....
0c0f0 17 00 49 00 4D 00 47 00-5F 00 32 00 30 00 32 00 ··I·M·G·_·2·0·2·
0c100 30 00 30 00 33 00 31 00-30 00 5F 00 31 00 33 00 0·0·3·1·0·_·1·3·
0c110 33 00 33 00 31 00 35 00-2E 00 6A 00 70 00 67 00 3·3·1·5·_·j·p·g·
```

Sel start = 49304, len = 136; clus = 786444; log sec = 6291552; phy sec = 6291584



MFT Entry: Data structure



Offset	Length	Data
0x00	6bytes	\$MFT record number of the parent directory
0x06	2bytes	Sequence number of the parent directory
0x08	8bytes	File creation time
0x10	8bytes	File modification time
0x18	8bytes	\$MFT modification time
0x20	8bytes	Last access time
0x28	8bytes	Allocated size of the index (if the record is an index)
0x30	8bytes	Actual size of the index (if the record is an index)
0x38	4bytes	File type flags
0x3c	4bytes	Reparse value
0x40	1byte	File name length in Unicode characters
0x41	1byte	File name type
0x42	Varies	File name



MFT Entry: Data structure



\$FILE_NAME Attribute Data Structure	
Offset	Length
0-7	File reference of parent directory
8-15	File creation time
16-23	File modification time
24-31	MFT modification time
32-39	File access time
40-47	Allocated size of file
48-55	Real size of file
56-59	Flags
60-63	Reparse value
64-64	Length of name
65-65	Namespace
66 +	Name



MFT Entry: Data structure



sFILE_NAME name space Field	
Name space value	Description
0	POSIX: The name is case sensitive and allows all Unicode characters except for '/' and NULL.
1	Win32: The name is case insensitive and allows most Unicode characters except for special values such as '/', '\', ':', '>', '<', and '?'.
2	DOS: The name is case insensitive, upper case, and no special characters. The name must have eight or fewer characters in the name and three or less in the extension.
3	Win32 & DOS: Used when the original name already fits in the DOS namespace and two names are not needed.



MFT Entry: Data structure



\$ATTRIBUTE_LIST Attribute Data Structure		
Offset	Length	Essential
0-3	Attribute type	Yes
4-5	Length of this entry	Yes
6-6	Length of name	Yes
7-7	Offset to name (relative to start of this entry)	Yes
8-15	Starting VCN in attribute	Yes
16-23	File reference where attribute is located	Yes
24-24	Attribute ID	Yes



MFT Entry: Data structure



\$OBJECT_ID Attribute Data Structure		
Offset	Length	Essential
0-15	Object ID	Yes
16-31	Birth Volume ID	No
32-47	Birth Object ID	No
48-63	Birth Domain ID	No



MFT Entry: Data structure



\$REPARSE_POINT Attribute Data Structure		
Offset	Length	Essential
0-3	Reparse type flags	Yes
4-5	Size of reparse data	Yes
6-7	Unused	No
8-9	Offset to target name (relative to byte 16)	Yes
10-11	Length of target name	Yes
12-13	Offset to print name of target (relative to byte 16)	Yes
14-15	Length of print name	Yes



MFT Entry: Data structure



\$INDEX_ROOT ATTRIBUTE Data Structure		
Offset	Length	Essential
0-3	Type of attribute in index (0 if entry does not use an attribute)	Yes
4-7	Collation sorting rule	Yes
8-11	Size of each index record in bytes	Yes
12-12	Size of each index record in clusters	Yes
13-15	Unused	No
16+	Node header	Yes



MFT Entry: Data structure



\$INDEX_ALLOCATION ATTRIBUTE Data Structure		
Offset	Length	Essential
0-3	Signature value ("INDX")	No
4-5	Offset to fixup array	Yes
6-7	Number of entries in fixup array	Yes
8-15	\$LogFile Sequence Number (LSN)	No
16-23	The VCN of this record in the full index stream	Yes
24+	Node header	Yes



MFT Entry: Data structure



\$BITMAP: Index Node Header Data Structure		
Offset	Length	Essential
0-3	Offset to start of index entry list (relative to start of the node header)	Yes
4-7	Offset to end of used portion of index entry list (relative to start of the node header)	Yes
8-11	Offset to end of allocated index entry list buffer (relative to start of the node header)	Yes
12-15	Flags	No



MFT Entry: Data structure



\$BITMAP: Generic Index Entry Data Structure	
Offset	Length
0-3	Undefined
4-5	Length of this entry
6-7	Length of content
8-15	Flags
16-23	Content



MFT Entry: Data structure



\$BITMAP: Generic Index Entry Data Structure	
Offset	Length
0-3	Undefined
4-5	Length of this entry



MFT Entry: Data structure



\$BITMAP: Directory Index Entry Data Structure		
Offset	Length	Essential
0-7	MFT file reference for file name	Yes
8-9	Length of this entry	Yes
10-11	Length of \$FILE_NAME attribute	No
12-15	Flags	Yes
16+	\$FILE_NAME Attribute (if length is > 0)	Yes
Last 8 bytes of entry, starting on an 8-byte boundary	VCN of child node in \$INDEX_ALLOCATION (field exists only if flag is set)	Yes



MFT Entry: Data structure



\$BOOT Data Structure		
Offset	Length	Essential
0-2	Assembly instruction to jump to boot code	No (unless it is the bootable file system)
3-10	OEM Name	No
11-12	Bytes per sector	Yes
13-13	Sectors per cluster	Yes
14-15	Reserved sectors (Microsoft says it must be 0)	No
16-20	Unused (Microsoft says it must be 0)	No
21-21	Media descriptor	No
22-23	Unused (Microsoft says it must be 0)	No



MFT Entry: Data structure



\$BOOT Data Structure		
Offset	Length	Essential
24-31	Unused (Microsoft says it is not checked)	No
32-35	Unused (Microsoft says it must be 0)	No
36-39	Unused (Microsoft says it is not checked)	No
40-47	Total sectors in file system	Yes
48-55	Starting cluster address of MFT	Yes
56-63	Starting cluster address of MFT Mirror \$DATA attribute	No
64-64	Size of file record (MFT entry)	Yes
65-67	Unused	No



MFT Entry: Data structure



\$BOOT Data Structure		
Offset	Length	Essential
68-68	Size of index record	Yes
69-71	Unused	No
72-79	Serial number	No
80-83	Unused	No
84-509	Boot code	No
510-511	Signature (oxaa55)	No



MFT Entry: Data structure



\$ATTRDEF FILE Data Structure		
Offset	Length	Essential
0-127	Name of attribute	Yes
128-131	Type identifier	Yes
132-135	Display rule	No
136-139	Collation rule	No
14-143	Flags	Yes
144-151	Minimum size	No
152-159	Maximum size	No



MFT Entry: Data structure



Flag values for the \$AttrDef entry	
Value	Description
0 x 02	Attribute can be used in an index
0 x 04	Attribute is always resident
0 x 08	Attribute can be non-resident



MFT Entry: Data structure



\$VOLUME FILE: \$VOLUME_INFORMATION Attribute Data Structure		
Offset	Length	Essential
0-7	Unused	No
8-8	Major version	Yes
9-9	Minor version	Yes
10-11	Flags	No



MFT Entry: Data structure



Flag values for the \$VOLUME_INFORMATION	
Value	Description
0x 00 01	Dirty
0 x 00 02	Resize \$LogFile (file system journal)
0 x 00 04	Upgrade volume next time
0 x 00 08	Mounted in NT
0 x 00 10	Deleting change journal
0 x 00 20	Repair object IDs
0 x 80 00	Modified by chkdsk



MFT Entry: Data structure



\$OBJID FILE Data Structure		
Offset	Length	Essential
0-1	Offset to file information	Yes
2-3	Size of file information	Yes
4-7	Unused	No
8-9	Size of index entry	Yes
10-11	Size of object ID (16-bytes)	Yes
12-15	Flags	Yes
16-31	Object ID	Yes
32-39	File Reference	Yes



MFT Entry: Data structure



\$OBJID FILE Data Structure		
Offset	Length	Essential
40-55	Birth volume ID	No
56-71	Birth object ID	No
72-87	Birth domain ID	No



MFT Entry: Data structure



\$QUOTA FILE Data Structure		
Offset	Length	Essential
0-1	Offset to owner ID (OFF)	Yes
2-3	Length of owner ID	Yes
4-7	Unused	No
8-9	Size of index entry	Yes
10-11	Size of SID (L)	Yes
12-15	Flags	Yes
16-(16+L-1)	SID	Yes
OFF+	Owner ID	Yes



MFT Entry: Data structure



Data structure for the \$Q index entries in \$Quota		
Offset	Length	Essential
0-1	Offset to quota information	Yes
2-3	Size of quota information	Yes
4-7	Unused	No
8-9	Size of index entry	Yes
10-11	Size of owner ID (4 bytes)	Yes
12-15	Flags	Yes
16-19	Owner ID	Yes
20-23	Version	No



MFT Entry: Data structure



Data structure for the \$Q index entries in \$Quota		
Offset	Length	Essential
24-27	Quota flags	Yes
28-35	Bytes charged to user	Yes
36-43	Time of last charge	No
44-51	Threshold value (a soft limit)	Yes
52-59	Hard limit value	Yes
60-67	Exceeded time	Yes
68-79	SID	Yes



MFT Entry: Data structure



Flag values for the \$Q index entry	
Flag	Description
0 x 00 00 00 01	Default limits being used
0 x 00 00 00 02	Limit reached
0 x 00 00 00 04	ID Deleted
0 x 00 00 00 10	Tracking data usage
0 x 00 00 00 20	Enforcing data usage
0 x 00 00 00 40	Usage tracking requested
0 x 00 00 00 80	Create log when threshold is met
0 x 00 00 01 00	Create log when limit is met
0 x 00 00 02 00	Out of date
0 x 00 00 04 00	Corrupt
0 x 00 00 08 00	Pending deletes



MFT Entry: Data structure



Data structure for the \$J attribute entries in \$UsrJrnl.		
Offset	Length	Essential
0-3	Size of this journal entry	Yes
4-5	Major version	Yes
6-7	Minor version	Yes
8-15	File reference of file that caused this entry	Yes
16-23	Parent directory file reference for file that caused this entry	No
24-31	USN for entry	Yes
32-39	Timestamp	Yes



MFT Entry: Data structure



Data structure for the \$J attribute entries in \$UsrJrnl		
Offset	Length	Essential
40-43	Flags for type of change	Yes
44-47	Source information	No
48-51	Security ID (SID)	No
52-55	File attributes	No
56-57	Size of file name	Yes
58+	File name	Yes



MFT Entry: Data structure



Values for the change type field in \$J entries	
Flag	Description
0 x 00 00 00 01	The default \$DATA attribute was overwritten
0 x 00 00 00 02	The default \$DATA attribute was extended
0 x 00 00 00 04	The default \$DATA attribute was truncated
0 x 00 00 00 10	A named \$DATA attribute was overwritten
0 x 00 00 00 20	A named \$DATA attribute was extended
0 x 00 00 00 40	A named \$DATA attribute was truncated
0 x 00 00 01 00	The file or directory was deleted
0 x 00 00 02 00	The file or directory was deleted
0 x 00 00 04 00	The extended attributes of the file were changed
0 x 00 00 08 00	The security descriptor was changed
0 x 00 00 10 00	The name changed—change journal entry has old name



MFT Entry: Data structure



Values for the change type field in \$J entries	
Flag	Description
0 x 00 00 20 00	The name changed—change journal entry has new name
0 x 00 00 40 00	Content indexed status changed
0 x 00 00 80 00	Changed basic file or directory attributes
0 x 00 01 00 00	A hard link was created or deleted
0 x 00 02 00 00	Compression status changed
0 x 00 04 00 00	Encryption status changed
0 x 00 08 00 00	Object ID changed
0 x 00 10 00 00	Repase point value changed
0 x 00 20 00 00	A named \$DATA attribute was created, deleted, or changed
0 x 80 00 00 00	The file or directory was closed



MFT Entry: Data structure



Data structure for the \$Max attribute of \$UsrJrnl		
Offset	Length	Essential
0-7	Maximum size	Yes
8-15	Allocation size	Yes
16-23	USN ID	Yes
24-31	Lowest USN	Yes



TASK



Attribute Analysis



References



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